Polynomial continued fractions with factorial reduction: hidden low-degree equivalences, a staged identification pipeline, and eight unidentified candidates

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Abstract

We search polynomial continued fractions (PCFs) with quadratic partial denominators and quartic partial numerators for full factorial reduction (FR) and remove every case explained by known mechanisms (the family of Conjecture 1.3 of Ben David et al., deflation, rational or quadratic limits, Möbius transforms of standard constants and of H. Cohen's databases, and Euler-type PCFs detected through hypergeometric solutions of the recurrence). One PCF that passed all numerical tests is shown to be of Euler type in disguise (Theorem 1). Fifteen PCFs remained whose order-four operator is irreducible over $\mathbb{Q}(z)$ and whose limits match none of 21,984 constants at 1,000 digits. A further search for Möbius equivalences with PCFs of LOWER degree shows that seven of the fifteen are disguised degree-$(1,2)$ PCFs, i.e. ratios of contiguous Gauss hypergeometric functions (agreement to 170 digits); for three of them we give exact proofs by polynomial coboundaries (Theorem 2), revealing a simple hiding mechanism that standard deflation tests do not detect. The remaining eight passed every test we could devise (Gamma products at 500 digits, 4,310 L-values of rational newforms, low-degree searches over about 10^9 partner PCFs, monodromy, parameter deformation) and are published as candidates, not as new constants. We release the data, the code and a staged identification-and-proof pipeline that would have identified the seven on the first day.

1 Conventions

A PCF is $x = a(0) + \dfrac{b(1)}{a(1) + \dfrac{b(2)}{a(2) + \cdots}}$ with $a, b \in \mathbb{Z}[n]$; P_n, Q_n satisfy $u_n = a(n)u_{n-1} + b(n)u_{n-2}$ with $P_{-1} = 1$, $P_0 = a(0)$, $Q_{-1} = 0$, $Q_0 = 1$. In matrix form $(u_{n-1}, u_n) = (u_{n-2}, u_{n-1})M(n)$ with $M(n) = \left(\begin{smallmatrix} 0 & b(n) \\ 1 & a(n) \end{smallmatrix}\right)$. Full FR means that $S_N = \frac{1}{N}\log(|Q_N|/\gcd(P_N, Q_N))$ stays bounded; numerically $|S_{800} - S_{200}| < 0.15$ (exact integers).

2 Search and filters

1. Brute force over about 2.3×10^9 PCFs ($\deg a = 2$, b a product of four linear factors) with Möbius hashing against standard constants: 239 unidentified FR-candidates, 45 with full FR, 23 outside Conjecture 1.3 and a ${}_3F_2$ scan.

2. Extended search on the six $a(n)$ of these 23 with $b(n) = s\prod_{i=1}^{4}(k_i n + t_i)$: 5,274 PCFs with full FR.

3. Removed: Conjecture 1.3 incl. index shifts (846), slow convergence (15), rational or quadratic limits (313), deflation (3,684); Möbius clustering left 347 new clusters; a Cohen-database limit (1), $\pi\sqrt{3}$ (1), ${}_3F_2$ (2).

4. Euler type (order-one right factor of the recurrence over $\mathbb{Q}(n)$, ore_algebra): 61 + 1 cases (Section 3).

5. $L = \theta^2(\theta-1)^2 - z\,a(\theta+1)\theta^2 - z^2\,b(\theta+2)$ annihilates $\sum_n Q_n z^n/(n!)^2$; fifteen operators are irreducible over $\mathbb{Q}(z)$ (labels C0–C14, SHA-256 time-stamped before disclosure).

3 A hidden Euler-type example

Theorem 1. *For $a(n) = 4n^2 + 10n - 3$, $b(n) = -n^2(n+4)(3n-4)$: $P_n = -(n+1)!\prod_{m=0}^{n-1}(3m+2)\,(n^2+3n+3)$ and $x = 1/(S-1/3)$ with $S = -\frac{1}{24}\sum_{k\ge1}\frac{k!\,(k+2)(k+3)(k+4)}{(2/3)_k\,3^k\,(k^2+k+1)(k^2+3k+3)}$.*

Proof sketch. $r(n) = P_n/P_{n-1} = (n+1)(3n-1)(n^2+3n+3)/(n^2+n+1)$ satisfies $a(n)r(n-1)+b(n) = r(n)r(n-1)$ in $\mathbb{Q}(n)$; the Casoratian of (P,Q) gives the series. Checked to 120 digits. □

With the same $a(n)$, $b(n) = (n+3)(2n-5)(2n-1)(3n-1)$ and $(n+1)(2n-3)(2n+1)(3n-5)$ give Möbius transforms of $\Gamma(1/3)^3\sqrt{3}/\pi$-type periods (250 digits); both operators factor as $[1,3]$.

4 Seven of the fifteen are disguised degree-$(1,2)$ PCFs

For each candidate x we searched PCFs y of lower degree (Batches 12–18: $\deg a \le 2$, $\deg b \le 2$, coefficient ranges up to ±60 for a and up to ±40 or factored with $k \le 12, |t| \le 30$ for b; about 10^9 partners) for a Möbius relation $y = (p+qx)/(r+sx)$, $|p|,\dots,|s| \le 8$, with planted, natural and random controls in every run. Every match was recomputed with exact recurrences to 160 digits.

cand.	partner $a(n)$	partner $b(n)$	relation	status
C0	$2n+3$	$(3n+5)(5n+1)$	$y = 3 - x$	proven
C10	$n-3$	$(2n+1)(3n+2)$	$y = (8-6x)/(4-5x)$	proven
C11	$n-3$	$(3n-1)(4n+1)$	$y = (8-3x)/(4+x)$	proven
C1	$2n-1$	$(3n-5)(5n-21)$	$y = -(5+4x)/5$	171 digits
C5	$6n-3$	$9(3n-2)(5n-2)$	$y = (x-3)/(-4-x)$	171 digits
C7	$2n$	$4(2n+3)(3n+1)$	$y = -3x/(5-x)$	171 digits
C14	$2n+8$	$4(4n+1)(5n-7)$	$y = (4+x)/6$	172 digits

Degree-$(1,2)$ PCFs are ratios of contiguous ${}_2F_1$ functions; hence none of the seven is a new constant.

Theorem 2. *Let M_1 be the matrix of candidate C0 ($a = 4n^2+10n-3$, $b = (2n-3)(2n+1)(3n+2)(5n-4)$) and M_2 that of its partner. Then $M_1(n)\,U(n) = (2n+1)\,U(n-1)\,M_2(n-1)$ with*

$$U(n) = \begin{pmatrix} \frac{10}{3}n^2 + \frac{7}{3}n - 2 & -10n^3 - \frac{41}{3}n^2 + 6n + \frac{5}{3} \\ -\frac{1}{3} & n \end{pmatrix},$$

$\det U(n) = -\frac{5}{9}(2n-1)(2n+1) \neq 0$ for integers n. Analogous identities hold for C10 ($\lambda = 3n+4$) and C11 ($\lambda = 3n+1$).

The identities are polynomial and were verified symbolically; they make the convergent matrices equivalent and imply the Möbius relations of the table. Mechanism: the candidate is its degree-$(1,2)$ partner shifted by one step and wrapped by one linear factor $\lambda(n)$ of its own $b(n)$, mixed by a cubic

polynomial gauge $U(n)$; the partner's b is then a product of two linear factors of the candidate's b, which turns the search into a cheap filter. C1, C5, C7 and C14 are not obtained this way within the searched gauge degrees (up to 5), nor by a two-step (contraction) ansatz whose positive control failed, so their mechanism is open.

5 The eight unidentified candidates

	$a(n)$	$b(n)$	x	ρ
C2	$3n^2+8n+8$	$2(2n+3)(3n+1)(3n+2)(3n+8)$	$33.5147685956496535\ldots$	$-187/42$
C3	$3n^2+8n+8$	$2(2n+3)(3n-1)(3n+8)(5n+3)$	$27.7112473245165813\ldots$	$-187/45$
C4	$3n^2-2n+2$	$2(2n-1)(3n-8)(3n-2)(5n-8)$	$-9.6178657191277317\ldots$	$3/5$
C6	$4n^2+9n+1$	$(2n+3)(4n+1)(4n+9)(6n+7)$	$39.4516159585089660\ldots$	$-123/28$
C8	$6n^2+13n+5$	$2(2n+1)(2n+5)(3n+4)(3n+7)$	$36.8847177941179069\ldots$	$-83/18$
C9	$6n^2+13n+5$	$(3n-10)(3n-4)(3n+10)(5n+3)$	$-41.3047171139817542\ldots$	$-121/60$
C12	$4n^2+10n-3$	$(2n-5)(2n+3)(3n+5)(5n-9)$	$49.6956732085298041\ldots$	$-39/20$
C13	$6n^2+13n+5$	$2(2n+5)(2n+9)(4n+3)(5n-8)$	$-54.0753445873853959\ldots$	$-241/52$

ρ: non-integer local exponent of L at its dominant singularity.

Tests passed by all eight (none identified). Not in Conjecture 1.3; not deflatable; limit not rational or quadratic; not Möbius ($|c| \le 10^5$) of 36 standard constants, of the 635 limits of arXiv:2409.06086v4, of evaluable limits of arXiv:2607.06581, or of the 171 euler2ai limits; no hypergeometric solution; operator irreducible; PSLQ $[1, x, c, xc]$ against 21,984 constants at 1,000 digits; no match in the OEIS; connection coefficients not Γ-products at 150 and 480 digits (planted controls recovered, incl. a 53-element basis); no Möbius relation ($|c| \le 10^8$) with 4,310 L-values of rational newforms of weight 2–4 and level ≤ 100 (300 digits); no low-degree Möbius partner in the searches of Section 4; traces of short words in the numerical monodromy are not algebraic of degree ≤ 4 (controls: hypergeometric operators give 100% algebraic traces); shifting one linear factor of b by its natural step gives no Möbius relation between neighbours of degree ≤ 4.

6 Limitations

Novelty is not established: unpublished databases (e.g. the Ramanujan Library) were not available to us. C12 was not completed by the widest mechanism filter. The identifications of C1, C5, C7, C14 are numerical (170 digits). Irreducibility is over $\mathbb{Q}(z)$ only. Full FR is a numerical criterion. The LMFDB has no search for decimal constants.

Data and code

All plans were time-stamped on GitHub (github.com/vichienFF/ramanujan-explorer) before the corresponding runs; scripts, logs, results and the pipeline `identify_pipeline.py` (stages: basic constants, known structures, mechanism filter, low-degree search, constant libraries, operator analysis, coboundary proof; self-test included) are released with this preprint.

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independent checker); all claims are reproducible with the released code.

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